2023 AMC 10B Problem 24 - Solution

Review the full statement and step-by-step solution for 2023 AMC 10B Problem 24. Great practice for AMC 10, AMC 12, AIME, and other math contests

2023 AMC 10B Problem 24

Problem:

What is the length of the boundary of the region in the xy plane consisting of points of the form $(2u-3w, v+4w)$ where $0 \leq u \leq 1, 0 \leq v \leq 1$, and $0 \leq w \leq 1$?

Answer Choices:

A. $10\sqrt{3}$

B. 13

C. 15

D. 18

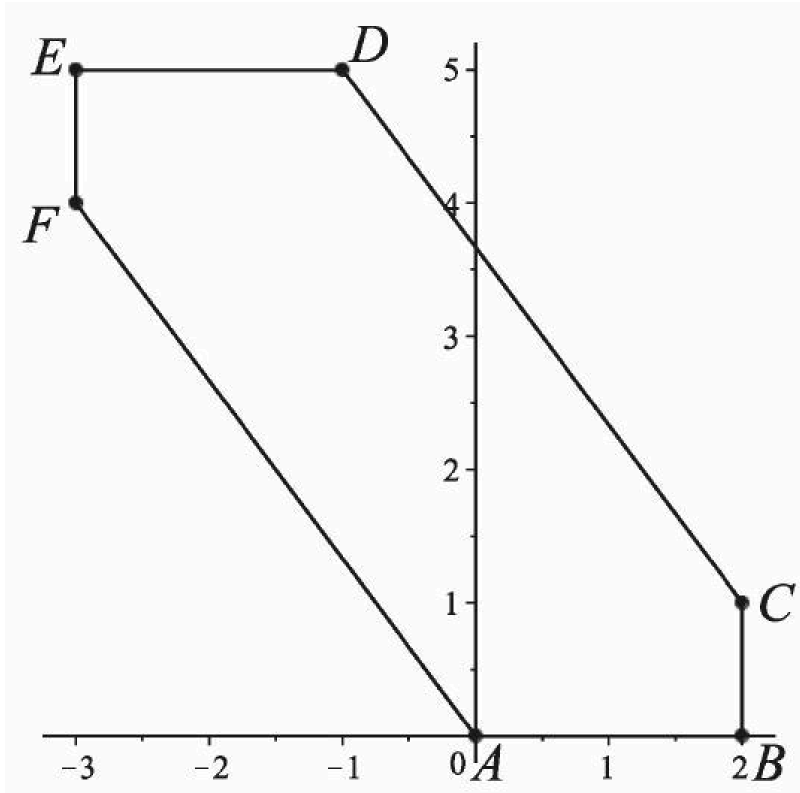
E. 16

Solution:

Call the region R. Let $f(u, v, w)=(2u-3w, v+4w)$. Then $A=(0,0)=f(0,0,0), B=(2,0)=f(1,0,0), C=(2,1)=f(1,1,0), D=(-1,5)=f(1,1,1), E=(-3,5)=\backslash$
$f(0,1,1)$, and $F=(-3,4)=f(0,0,1)$ are points in R. Note that if $(x_1, y_1)=f(u_1, v_1, w_1)$ and $(x_2, y_2)=f(u_2, v_2, w_2)$, then

$$t(x_1, y_1)+(1-t)(x_2, y_2)=f(tu_1+(1-t)u_2, tv_1+(1-t)v_2, tw_1+(1-t)w_2)$$

for all real numbers t. If $0 \leq t \leq 1$, then $tu_1 + (1-t)u_2$ lies between u_1 and u_2. Thus if also $0 \leq u_1 \leq 1$ and $0 \leq u_2 \leq 1$, then $0 \leq tu_1 + (1-t)u_2 \leq 1$, and similarly for the v_i and w_i. Thus if $\boldsymbol{p}_1 = (x_1, y_1)$ and $\boldsymbol{p}_2 = (x_2, y_2)$ are points of R that arise from triples (u_1, v_1, w_1) and (u_2, v_2, w_2) in the unit cube $[0,1]^3$, then the line segment $t\boldsymbol{p}_1 + (1-t)\boldsymbol{p}_2, 0 \leq t \leq 1$, joining them also lies entirely in R. Hence the edges of the hexagon $ABCDEF$ all lie in R, and thus also the interior of this hexagon.



To see that there are no other points in R, note that because $x = 2u - 3w$, it follows that $-3 \leq x \leq 2$. The segments $\overline{BC}$ and $\overline{EF}$ lie on the boundary of this strip. Also, because $y = v + 4w$, it follows that $0 \leq y \leq 5$. The segments $\overline{AB}$ and $\overline{DE}$ lie on the boundary of this strip. Finally, because $4x + 3y = 4(2u - 3w) + 3(v + 4w) = 8u + 3v$, it follows that $0 \leq 4x + 3y \leq 11$. The segments $\overline{CD}$ and $\overline{AF}$ lie on the boundary of this strip. Because all points of R must lie in the intersection of these three strips, all points of R have been identified.\
The requested length of the boundary of R is the perimeter of the hexagon:

$$AB + BC + CD + DE + EF + FA = 2 + 1 + \sqrt{3^2 + 4^2} + 2 + 1 + \sqrt{3^2 + 4^2} = \text{(E)}\boxed{16}.$$

Note: The interval $[0,1]$ on the real line is convex, so the unit cube $[0,1]^3$ is convex. The image of a convex set under a linear transformation is convex. Also, a convex set in the plane can be described by its supporting half-planes. This is particularly simple when the set in question is a convex polygon.